



FXD Engineering Project Binder

Volume 2 - Milestones 26-31

Audited engineering record based on the approved documentation-recovery structure

Project	FXD - Intelligent Industrial Fixture Design
Repository	kool1160/fxd-fixture-design
Document Class	Engineering development history / professional project binder
Volume Status	Closed audited record through Milestone 31
Audit Basis	Merged pull requests, tests, CI, project law, architecture, decision, and project records
Runtime Changes	None - documentation only

AI proposes. Engineering validates.

The software should think like an experienced fixture engineer.

1. Executive Summary

Volume 2 records the transformation of FXD from a completed deterministic fixture-engineering foundation into a serious local engineering application with real STEP viewing, a unified PySide6/VTK workbench, interactive engineering workflows, controlled branding, real fixture-construction geometry, and the advisory AI Fixture Engineer with guided validation.

Governing principles

- Source customer CAD remains immutable.
- Deterministic geometry, mathematics, and engineering rules remain authoritative.
- AI is advisory and may not approve a fixture.
- Every recommendation and validation result must remain explainable and traceable.
- Manufacturing practicality and qualified human approval remain mandatory.

2. Master Milestone Recovery Tracker

Exact historical test counts are reported only when the controlling repository evidence states them.

M	Milestone	Status / PR	Scope Summary	Testing / Evidence	Boundary
26	Complete the Local Engineering Workbench	Complete #48	Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.	5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.	Render benchmarks were not visible-GUI FPS claims; engineering review only.
27	Unify the FXD Engineering Workbench	Complete #49	Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.	15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.	Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.
28	Expose the Interactive Fixture Engineering Workflow	Complete #50	Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.	34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.	Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.
29	Implement the Desktop UI and Branding System	Complete #51	Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.	30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.	Branding does not alter engineering rules, source identity, validation, or approval authority.
30	Generate Real Fixture Geometry and Tack/Location Workflows	Complete #52	Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.	16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.	No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release.
31	AI Fixture Engineer and Guided Validation	Complete #53	Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence.	17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed.	AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.

3. Detailed Milestone Records

Milestone 26 - Complete the Local Engineering Workbench

Status	Complete	Controlling PR	#48
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Engineering scope and significance

Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.

Validation and source evidence

5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.

Known limitations and protected boundaries

Render benchmarks were not visible-GUI FPS claims; engineering review only.

Completion record

Milestone 26 is recorded as Complete because its controlling implementation was merged through PR #48. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 27 - Unify the FXD Engineering Workbench

Status	Complete	Controlling PR	#49
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Engineering scope and significance

Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.

Validation and source evidence

15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.

Known limitations and protected boundaries

Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.

Completion record

Milestone 27 is recorded as Complete because its controlling implementation was merged through PR #49. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 28 - Expose the Interactive Fixture Engineering Workflow

Status	Complete	Controlling PR	#50
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Engineering scope and significance

Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.

Validation and source evidence

34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.

Known limitations and protected boundaries

Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.

Completion record

Milestone 28 is recorded as Complete because its controlling implementation was merged through PR #50. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 29 - Implement the Desktop UI and Branding System

Status	Complete	Controlling PR	#51
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Engineering scope and significance

Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.

Validation and source evidence

30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.

Known limitations and protected boundaries

Branding does not alter engineering rules, source identity, validation, or approval authority.

Completion record

Milestone 29 is recorded as Complete because its controlling implementation was merged through PR #51. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 30 - Generate Real Fixture Geometry and Tack/Location Workflows

Status	Complete	Controlling PR	#52
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Engineering scope and significance

Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.

Validation and source evidence

16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.

Known limitations and protected boundaries

No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release.

Completion record

Milestone 30 is recorded as Complete because its controlling implementation was merged through PR #52. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

Milestone 31 - AI Fixture Engineer and Guided Validation

Status	Complete	Controlling PR	#53
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Engineering scope and significance

Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence.

Validation and source evidence

17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed.

Known limitations and protected boundaries

AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.

Completion record

Milestone 31 is recorded as Complete because its controlling implementation was merged through PR #53. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

4. Evidence and Validation Record

Controlling evidence consists of merged pull requests, commit history, automated and hosted validation, real-kernel proof, Windows visual evidence where applicable, architecture decisions, project records, roadmap records, and explicit human-review boundaries. Exact missing historical values are not invented.

5. Documentation Correction Note

This audited edition replaces the earlier under-filled print layout. It restores dense milestone tracking, engineering scope, validation evidence, limitations, closeout, and content-driven pagination while preserving repository history as the source of truth.

6. Volume Closeout

Volume 2 closes at Milestone 31. FXD now has a professional Windows engineering workbench, real source-geometry review, interactive intent and correction workflows, deterministic real fixture-build geometry, governed persistence, and an advisory AI proposal layer subordinate to validation and human authority.

7. Controlled Handoff

Work after Milestone 31 belongs to a later binder volume or addendum. Its current status is governed outside this retrospective binder by the authoritative milestone registry and active GitHub issue.

8. Future Documentation Rule

- Planning -> Implementation -> Testing -> Documentation.
- Never mark an open or unmerged milestone complete.
- Update controlled binder batches while evidence is fresh.
- Record PR, validation evidence, limitations, and the next handoff.
- Pass results, not noise; never reconstruct missing facts by guessing.